

Dr. Sergio Ruocco

Embedded and real-time systems, Linux and toolchain consultant · Senior Software Engineer at EMASS
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Profile

Dr. Sergio Ruocco is an embedded-systems and toolchain consultant with 30 years of experience in Linux kernel and driver development, real-time operating systems, compiler toolchains, and non-volatile memory. He helps teams de-risk unfamiliar hardware and software: porting operating systems and SDKs to new SoCs, writing and debugging kernel drivers, building GCC backends, improving real-time behaviour, analysing system performance, and setting up CI/CD build and test pipelines and IT/DevOps infrastructure. He is currently a Senior Software Engineer at EMASS, an NTU/A*STAR spin-off, where he builds the full software stack and the CI/CD and IT/DevOps infrastructure for the ECS-DoT ultra-low-power RISC-V AI SoC, and planned and delivered its Zephyr RTOS port, mentoring a new team. His clients have included Seagate (USA) and Micron Italy, and he has led EU ENIAC projects with STMicroelectronics.

Capabilities

Engagements range from architecture reviews and feasibility studies to hands-on porting, driver development and debugging.

Operating systems and platforms. Linux (Debian, Ubuntu, Mint, Red Hat) on workstations and datacentre servers; L4 microkernels, MbedOS, Zephyr, RTOS internals and porting; Linux kernel, memory management and device drivers; ZFS and zfs-root workstations and storage servers; IPMI, RAID, GPU servers; Secure build systems and export-controlled software licence servers.

Languages and toolchains. C, GCC/G++, GDB and the GNU toolchain, native and cross-compiled; GCC backend and cross-compiler development; Buildroot, Android SDK, MbedOS SDK; Python, Bash, Git, SSH, GPG, tmux/screen; AI-assisted coding; Docker and Buildx, GitLab CI multi-project pipelines, GitHub and GitLab container registries; IT/DevOps infrastructure for engineering teams.

Architectures and hardware. ARM and RISC-V platforms; board bring-up, JTAG and RISC-V debug (OpenOCD); Microcontroller firmware on RP2040/RP2350: Pico SDK, PIO, TinyUSB, I²C and UART; FPGA boards, peripherals, memory maps and I/O devices; Non-volatile memory (NVM-DIMM, PCM, STT-MRAM) and storage systems; AI accelerators and video-compression hardware.

Methods. Real-time scheduling and temporal reflection; Performance analysis and system tracing; Debugging poorly documented hardware and software; Technical leadership, mentoring, procurement and project delivery.

Selected projects

Full software stack for the ECS-DoT ultra-low-power RISC-V AI SoC

*2024 – present · NTU, then EMASS (an NTU/A*STAR spin-off) from 06/2025*

Research and develop the device drivers, RTOS, SDK and toolchains needed to compile, optimise and run AI applications on ECS-DoT, a custom ultra-low-power SoC with a RISC-V core and dedicated hardware accelerators, designed and fabricated by the team.

Impact: Enables AI workloads on the ECS-DoT SoC; builds on the AI-accelerator driver and RTOS work done at A*STAR I²R.

ECS-DoT boot agent and JTAG adapter on the Raspberry Pi Pico 2

2026 – present · EMASS · ECS-DoT development boards

Extended the open-source DirtyJTAG firmware into the boards' boot agent: it loads RISC-V applications into the SoC over JTAG at power-up, serves as an OpenOCD adapter and USB-serial bridge, and drives the boards' power control and measurement.

Impact: Shipped to customers in tagged firmware releases since 07/2026.

MbedOS and SDK port to the ECS-DoT RISC-V SoC

*2020 – 2024 · A*STAR I²R / NTU · ECS-DoT Edge SoC with AI acceleration*

Ported the ARM MbedOS 5.12.4 RTOS and its SDK to the ECS-DoT SoC, covering a different CPU core, instruction set, peripherals, memory map and I/O devices. Both the original and the target platforms were poorly documented and buggy, so the work was a complex trial-and-error effort with the GCC toolchain and GDB.

Impact: Delivered a working RTOS port and board-support package; technology disclosure SWIP-PMEP-2022-31, licensed to EMASS Pte Ltd.

AI accelerator Linux kernel driver port to MbedOS on RISC-V

*2022 – 2023 · A*STAR I²R · ECS-DoT*

Ported the AI accelerator's ARM Linux kernel driver to MbedOS on the ECS-DoT RISC-V CPU, adapting the driver to a new operating system, memory map and device model.

Impact: Technology disclosure SWIP-PNVD-2022-33, licensed to EMASS Pte Ltd.

Real-time extensions and reflective scheduling for the L4 microkernel

2004 – 2007 · NICTA · ERTOS Programme

Analysed the real-time behaviour of general-purpose L4 microkernels, identified the architectural obstacles to predictable scheduling, and designed and implemented a reflective real-time scheduler that lets a system observe and control its own temporal behaviour.

Impact: Published at IEEE RTSS 2006 and OSPERT 2006/2007 and in the EURASIP Journal on Embedded Systems; the work influenced the real-time features of the commercial OKL4 microkernel.

Linux memory-management extensions for non-volatile DIMMs

*2015 · A*STAR DSI · CERN OpenLab collaboration*

Extended Linux memory management to support non-volatile DIMMs, allowing persistent data to be placed directly in main memory and processed at memory speed.

Impact: Procured an invitation to collaborate with CERN OpenLab and an invited talk at CERN OpenLab 2015.

GCC backend for a custom 16-bit microcontroller

2009 – 2010 · Micron Italy (formerly STMicroelectronics) · Università di Milano-Bicocca

Ported the GNU GCC C compiler to a custom, non-standard 16-bit microcontroller managing embedded NAND/NOR phase-change memories under hard real-time constraints and security requirements. The target was poorly documented, requiring a new backend and extensive debugging with GDB.

Impact: Delivered under a Micron R&D contract; the compiler supported devices deployed in high volume.

Active Discs Clusters Performance Study (ADCPS)

*2017 – 2019 · A*STAR DSI, then I²R · funded by Seagate USA*

Industry-funded performance study of active disc clusters, covering workload characterisation, measurement and analysis.

Impact: Delivered a performance study to Seagate USA.

EU ENIAC SMART and TOISE projects

2009 – 2013 · Università di Milano-Bicocca · Micron Italy and STMicroelectronics

Principal Investigator and technical lead for UNIMIB in two three-year EU ENIAC projects: SMART (techniques and tools for advanced real-time embedded firmware, €302,400) and TOISE (trusted computing for European embedded systems, €556,000). Responsible for planning, budget, recruiting research engineers and post-docs, and scientific and technical delivery.

Impact: Funded a research fellowship and a team of post-docs and research engineers; produced tools and techniques for advanced embedded firmware.

Experience

Senior Software Engineer — EMASS (an NTU/A*STAR spin-off)

06/2025 – present · Singapore

End-to-end software for ECS-DoT, an ultra-low-power AI SoC with a RISC-V core and dedicated hardware accelerators. Research and development of the full stack of software and tools — device drivers, RTOS, SDK and toolchains — to compile, optimise and run AI applications on a custom SoC designed and fabricated by the team.

Research Fellow — School of Computer Science and Engineering, Nanyang Technological University (NTU)

04/2024 – 05/2025 · Singapore

Kept the ECS-DoT system-software engineering going between A*STAR I²R and EMASS, laying the groundwork for the EMASS role: the first bare-metal SDK, its SPI and camera drivers, and analysis of hardware bugs in the SoC's SPI peripheral.

Senior Scientist II — A*STAR Institute for Infocomm Research (I²R)

06/2018 – 09/2023 · Singapore

Embedded real-time operating systems, SDKs, firmware and drivers for a RISC-V-based SoC customised with a deep-learning accelerator for ultra-low-power edge-IoT applications.

Scientist II — A*STAR Data Storage Institute (DSI)

10/2012 – 05/2018 · Singapore

Systems software and programming-toolchain research for next-generation solid-state storage and non-volatile memory.

Research Fellow — Dipartimento di Informatica, Sistemistica e Comunicazione (DISCo), Università di Milano-Bicocca
04/2007 – 09/2012 · Milan, Italy
Research and teaching in real-time embedded systems, compiler retargeting and toolchains, and next-generation non-volatile storage, with projects funded by industry contracts and European research grants.

Researcher, Embedded, Real-Time and Operating Systems (ERTOS) Programme — NICTA (National ICT Australia, now Data61/CSIRO), Sydney
04/2004 – 03/2007 · Australia
Real-time and operating-systems research on the L4 microkernel.

Linux Kernel Engineer — Comerson s.r.l., Milan
2002 – 2004 · Italy
Smart video camera with TCP/IP support — at a time when other digital webcams were serial or USB.

Software Engineer — Metamatica, Milan
2000 – 2001 · Italy
Web-based lottery game.

Embedded Software Engineer — CD Express
1996 – 1997 · Italy
MPEG decoder for the Commodore CD-32.

Software Engineer and Independent Consultant — Self-employed
01/1990 – 01/2002 · Italy
Consulting on embedded real-time systems while attending university.

Patents and technology disclosures

*EMASS (Embedded A.I. Systems Pte Ltd) is a Singaporean fabless semiconductor company founded in 2020 as a spin-off from NTU Singapore and A*STAR research, and acquired by Nanoveu Ltd (ASX: NVU) in March 2025.*

SWIP-PMEP-2022-31 — Porting of MbedOS 5.12.4 to the ECS-DoT SoC and RISC-V CPU. Filed 2023-03-10. Review completed 27 April 2023. Licensed to EMASS Pte Ltd.

SWIP-PNVD-2022-33 — Porting of the AI accelerator ARM Linux kernel driver for MbedOS on the ECS-DoT RISC-V CPU. Filed 2023-03-10. Review completed 2 May 2023. Licensed to EMASS Pte Ltd.

SWIP-TLCN-2022-32 — Tracing Linux compiler and kernel driver for an AI accelerator. Filed 2023-03-10. Review completed 21 April 2023. Licensed to EMASS Pte Ltd.

TD2022128 / I2R/Z/13931 — Edge SoC with AI acceleration. Filed 2023-01-12. Licensed to EMASS Pte Ltd.

Publications

Selected publications.

1. Q. Trinh, S. Ruocco, M. Alioto, “Time-Based Sensing for Reference-Less and Robust Read in STT-MRAM Memories,” *IEEE Transactions on Circuits and Systems I: Regular Papers*, 65(10):3338–3348. DOI
2. Q. Trinh, S. Ruocco, M. Alioto, “Novel Time-Based Sensing Scheme for STT-MRAMs,” *IEEE International Symposium on Circuits and Systems (ISCAS)*, Florence, Italy, 1–5. DOI
3. S. Ruocco, D.-K. Le, “Efficient Persistence of Financial Transactions in NVM-based Cloud Data Centers,” *International Conference on Cloud Computing Research and Innovation (ICCCRI)*, 25–36. DOI
4. S. Ruocco, “A Real-Time Programmer’s Tour of General-Purpose L4 Microkernels,” *EURASIP Journal on Embedded Systems*, 2008:1–14. DOI
5. S. Ruocco, “User-Level Fine-Grained Adaptive Real-Time Scheduling via Temporal Reflection,” *27th IEEE Real-Time Systems Symposium (RTSS)*, Rio de Janeiro, Brazil, 246–256. DOI

Education

2004 — PhD in Computer Science
Università degli Studi di Milano, Italy — Mathematics, Statistics, Computational Sciences and Computer Science Research
Thesis: *Temporal Reflection*. Doctoral scholarship, 2000–2004. In Italy, PhD degrees are pass/no-pass and are not graded.

1999 — Master of Science in Computer Science
Università degli Studi di Milano, Italy — 110/110 cum laude
Thesis: *HyperReal: a paradigm for the design and synthesis of hard real-time control systems*.